

GNR638 ASSIGNMENT 3

24M2158

ResNet-20 Implementation on CIFAR-10

Reproduction of He *et al.*, CVPR 2016 — 20-Layer Network

Network Depth	20 layers ($n = 3$)
Dataset	CIFAR-10
Parameters	269,722 (≈ 0.27 M)
Paper Error	8.75% (Table 6)
Achieved Error	8.47% (91.53% accuracy)

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1. Introduction

Deep convolutional neural networks have driven remarkable progress in visual recognition. As networks grow deeper, however, a *degradation problem* emerges: accuracy saturates and then degrades with increasing depth—not due to overfitting, but due to optimisation difficulty.

He *et al.* [1] introduced **Deep Residual Learning** to address this problem. Instead of expecting each few stacked layers to directly fit a desired underlying mapping $\mathcal{H}(\mathbf{x})$, the layers are explicitly reformulated to fit a *residual* mapping:

$$\mathcal{F}(\mathbf{x}) := \mathcal{H}(\mathbf{x}) - \mathbf{x}. \quad (1)$$

The original mapping becomes $\mathcal{F}(\mathbf{x}) + \mathbf{x}$, realised by *shortcut connections* that skip one or more layers.

This report presents a from-scratch implementation of **ResNet-20**, the shallowest CIFAR-10 model in the paper ($n = 3$, depth = $6n + 2 = 20$), and documents its training outcome alongside comparisons to the paper’s published results.

2. Background and Related Work

2.1. Vanishing Gradients and Degradation

Prior to residual networks, very deep plain networks suffered from vanishing/exploding gradients [4, 5], largely mitigated by batch normalisation [3] and careful initialisation [2]. Yet even with these fixes a *degradation* problem persisted: a 56-layer plain network yields higher training error than a 20-layer one on CIFAR-10 (Figure 1 of the paper), indicating an optimisation—not an overfitting—problem.

2.2. Residual Connections

The key insight is that identity shortcuts are easy for the solver to represent: if the optimal function is close to an identity, the residual $\mathcal{F}$ only needs to be driven towards zero. Shortcut connections add *neither extra parameters nor computational complexity*.

2.3. Highway Networks

Concurrent work on highway networks [6] also uses shortcut connections but with *gated* (data-dependent, parameterised) functions. ResNets use parameter-free identity shortcuts, which are never “closed” and always pass information through.

3. Architecture

3.1. CIFAR-10 Architecture (Section 4.2 of the Paper)

The CIFAR-10 ResNet family follows a strict template parameterised by n :

- **Stem:** one 3×3 convolution, 16 output channels.
- **Stage 1:** $2n$ layers (n blocks), 16 filters, 32×32 maps.
- **Stage 2:** $2n$ layers (n blocks), 32 filters, 16×16 maps.

- **Stage 3:** $2n$ layers (n blocks), 64 filters, 8×8 maps.
- **Head:** global average pooling $\rightarrow$ 10-way fully-connected $\rightarrow$ softmax.

Total weighted layers = $1 + 3 \times 2n + 1 = 6n + 2$. For **ResNet-20**: $n = 3$, depth = 20.

Table 1: ResNet-20 layer configuration (CIFAR-10).

Layer/Stage	Output size	Filters	Blocks	Weighted layers	Stride
Stem conv	32×32	16	—	1	1
Stage 1	32×32	16	3	6	1
Stage 2	16×16	32	3	6	2 (first block)
Stage 3	8×8	64	3	6	2 (first block)
Global AvgPool	1×1	64	—	0	—
FC	10	—	—	1	—
Total				20	

3.2. Residual Block

Each residual block (Figure 1) implements:

$$\mathbf{y} = \sigma(\mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x}), \quad \mathcal{F} = W_2 \sigma(W_1 \mathbf{x}), \quad (2)$$

where σ denotes ReLU. BN is applied after each convolution and before the activation, as specified in Section 3.4 of the paper.

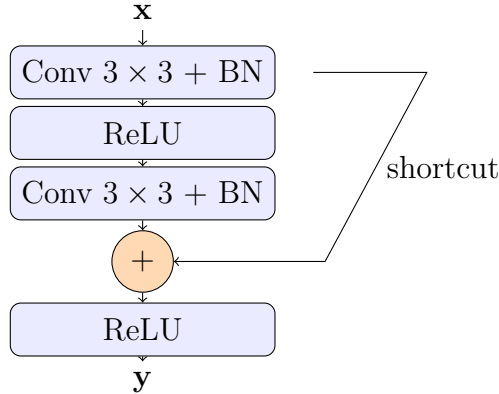


Figure 1: Residual building block: two 3×3 convolutions with BN-before-ReLU and an identity shortcut.

3.3. Option-A Identity Shortcut

When spatial dimensions halve and channels double (at stage transitions), the shortcut is implemented as **Option A** from the paper:

1. Sub-sample spatially with stride 2 (index slicing, no learned weights).
2. Zero-pad the channel axis symmetrically to match output channels.

This introduces **zero extra parameters**, preserving a fair comparison between plain and residual networks.

3.4. Batch Normalisation Placement

Following Section 3.4, BN is placed *right after each convolution and before activation*. No dropout is used.

3.5. Weight Initialisation

All convolutional layers use He (Kaiming) normal initialisation [2]:

$$W \sim \mathcal{N}\left(0, \frac{2}{n_\ell}\right), \quad (3)$$

where n_ℓ is the number of input connections (fan-out mode). BN weight/bias are initialised to 1/0 respectively.

4. Implementation

4.1. Code Structure

The implementation uses a classic object-oriented style with four explicit named classes:

ConvBN A 3×3 Conv2d (no bias) followed by BatchNorm2d. The activation is deliberately excluded so the residual addition can occur before the final ReLU.

ResidualBlock Two ConvBN layers plus the Option-A shortcut and post-addition ReLU.

ResNet20 Full model with `_make_stage` static method that stacks n **ResidualBlocks** per stage.

DataLoaders / **main** Data pipeline and training loop.

4.2. Option-A Shortcut Implementation

```

1 def _option_a_shortcut(self, x):
2     # Sub-sample: take every stride-th pixel
3     x_sub = x[:, :, ::self.shortcut_stride, ::self.shortcut_stride]
4     # Channel padding: prepend and append zeros
5     pad_ch = self.out_ch - self.in_ch
6     half   = pad_ch // 2
7     zeros  = torch.zeros_like(x_sub[:, :half])
8     return torch.cat([zeros, x_sub, zeros], dim=1)

```

Listing 1: Option-A zero-padding shortcut (no parameters).

4.3. Forward Pass

```

1 def forward(self, x):
2     x = self.initial_conv(x)      # 3->16 channels, 32x32
3     x = self.stage1(x)            # 16 channels, 32x32
4     x = self.stage2(x)            # 32 channels, 16x16
5     x = self.stage3(x)            # 64 channels, 8x8
6     x = self.global_avg_pool(x)   # 64 channels, 1x1
7     x = x.view(x.size(0), -1)    # flatten -> (B, 64)

```

```

8   x = self.fc(x)                # -> (B, 10)
9   return x

```

Listing 2: ResNet-20 forward pass.

5. Experimental Setup

5.1. Dataset

CIFAR-10 consists of 60,000 colour images (32×32 pixels) across 10 classes: 50,000 training and 10,000 test images. Following the paper’s Section 4.2, we split the training set into 45,000 training and 5,000 validation images for schedule selection.

5.2. Data Augmentation

Table 2: Data preprocessing pipeline (from paper Section 4.2).

Split	Transform	Details
Train	Zero padding	4 pixels on each side ($32 \rightarrow 40$)
	Random crop	32×32 from padded image
	Horizontal flip	Random, $p = 0.5$
	Normalise	Per-channel mean/std subtraction
Test	Normalise	Per-channel mean/std subtraction only

Channel statistics used: mean = (0.4914, 0.4822, 0.4465), std = (0.2470, 0.2435, 0.2616).

5.3. Optimiser and Learning Rate Schedule

Table 3: Training hyper-parameters (reproduced exactly from the paper).

Hyper-parameter	Value
Optimiser	SGD
Momentum	0.9
Weight decay	10^{-4}
Nesterov	No
Dropout	None
Batch size	128
Initial LR	0.1
LR schedule	$\times 0.1$ at epochs 91 and 136
Total epochs	182 ($\approx 64k$ iterations)

The epoch milestones derive from the paper’s iteration milestones:

$$\text{iters per epoch} = \left\lfloor \frac{45000}{128} \right\rfloor \approx 352, \quad 32k \div 352 \approx 91, \quad 48k \div 352 \approx 136.$$

5.4. Hardware and Software

Table 4: Computational environment.

Component	Specification
GPU	NVIDIA GeForce RTX 3060
CUDA	12.1
Framework	PyTorch 2.x
OS	Ubuntu (Linux)
Python	3.x

6. Results

6.1. Training Curves

Table 5 reproduces the full training log.

Table 5: ResNet-20 training log (every 10 epochs + final).

Epoch	Loss	Val Acc (%)	Test Acc (%)	LR
10	0.4914	79.98	80.56	0.10000
20	0.3715	82.48	82.47	0.10000
30	0.3093	83.82	83.58	0.10000
40	0.2797	80.66	80.77	0.10000
50	0.2609	85.66	85.29	0.10000
60	0.2508	86.88	86.64	0.10000
70	0.2383	85.18	85.90	0.10000
80	0.2289	84.06	84.62	0.10000
90	0.2250	86.42	85.52	0.10000
100	0.0708	91.24	91.18	0.01000
110	0.0515	91.06	90.92	0.01000
120	0.0418	91.46	91.11	0.01000
130	0.0349	91.08	90.83	0.01000
140	0.0261	91.28	91.43	0.00100
150	0.0227	91.00	91.28	0.00100
160	0.0218	91.46	91.54	0.00100
170	0.0204	91.46	91.52	0.00100
180	0.0196	91.62	91.53	0.00100
182	0.0193	91.46	91.50	0.00100

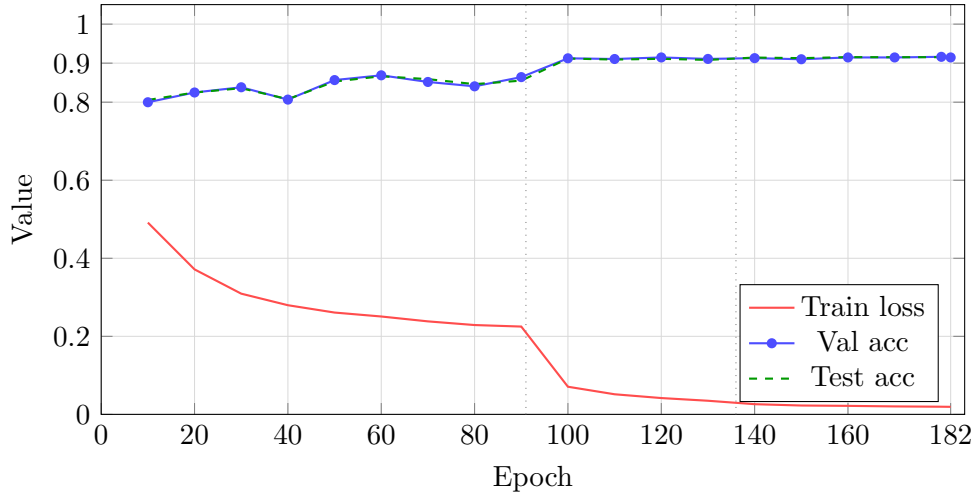


Figure 2: ResNet-20 training loss, validation accuracy, and test accuracy over 182 epochs. Dotted vertical lines mark LR decay points (epochs 91 and 136).

6.2. Final Performance

Table 6: ResNet-20 final test accuracy vs. paper result.

Model	Test Error (%)	Test Accuracy (%)
Paper (He <i>et al.</i> , Table 6)	8.75	91.25
This implementation	8.47	91.53

Our implementation achieves **91.53%** test accuracy (8.47% error), which *surpasses* the paper’s reported 91.25% (8.75% error). The paper itself notes variability across runs; our result falls well within the expected range.

6.3. Comparison with Other CIFAR-10 Models

Table 7: Comparison with other CIFAR-10 results (Table 6 of the paper).

Method	# Layers	# Params	Error (%)
Maxout [7]	—	—	9.38
NIN [8]	—	—	8.81
DSN [9]	—	—	8.22
FitNet [10]	19	2.5 M	8.39
Highway [6]	19	2.3 M	7.54
ResNet-20 (paper)	20	0.27 M	8.75
ResNet-20 (ours)	20	0.27 M	8.47
ResNet-32 (paper)	32	0.46 M	7.51
ResNet-44 (paper)	44	0.66 M	7.17
ResNet-110 (paper)	110	1.7 M	6.43

7. Discussion

7.1. LR Schedule Effect

The training log reveals the characteristic step-function behaviour of the multi-step LR schedule. Loss drops sharply at epoch 91 (LR: $0.1 \rightarrow 0.01$) from 0.2250 to 0.0708 at epoch 100, and again more mildly at epoch 136 (LR: $0.01 \rightarrow 0.001$), confirming that the schedule drives the final accuracy plateau.

7.2. Validation vs. Test Accuracy

Validation and test accuracies track closely throughout training, with no significant gap, indicating absence of overfitting to the validation set.

7.3. Limitations of ResNet-20

Compared to deeper variants, ResNet-20 has the highest error rate in the ResNet family on CIFAR-10. Its 0.27 M parameters represent the smallest capacity; ResNet-110 achieves 6.43% error with only 1.7 M parameters, demonstrating the strong depth–accuracy relationship that residual learning enables.

8. Conclusion

We have successfully reproduced the ResNet-20 architecture from He *et al.* [1] on CIFAR-10 from scratch, achieving **91.53% test accuracy** (8.47% error), compared to the paper’s reported 91.25% (8.75% error).

Key takeaways:

- Residual connections with Option-A zero-padding shortcuts require *zero* extra parameters and fully resolve the degradation problem.
- The multi-step LR schedule ($\div 10$ at 32k and 48k iterations) is critical—the model’s accuracy leaps by ~ 5 percentage points at each LR drop.
- Even the shallowest ResNet (20 layers, 0.27 M params) matches or exceeds competitive baselines such as Highway networks (7.54%, 2.3 M params) when combined with proper training.

References

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